

Benchmarking TCR–Epitope Recognition Models across Antigen Mutations

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Abstract

Motivation: T cell receptor (TCR)–epitope prediction models are commonly evaluated on collections of positive and negative TCR–peptide pairs or on generalization to unseen epitopes. These settings do not directly test whether models can reproduce experimentally observed changes in TCR recognition caused by controlled antigen sequence perturbations.

Results: We introduce a mutation-aware evaluation framework in which all test peptides are single-amino-acid variants of a common reference antigen and prediction performance is assessed against experimentally measured retention or loss of TCR recognition. Using the HLA-A*02:01-restricted SARS-CoV-2 epitope YLQPRTFLL, we constructed a benchmark containing all 171 possible single-amino-acid substitutions evaluated across 21 TCRs, yielding 3,591 TCR–mutant interactions. Eight representative prediction models were assessed using mutation-specific standardized partial ROC-AUC at FPR ≤ 0.1 . Overall performance was modest, with the best Macro AUC_{0.1} below 0.65, and prediction quality varied substantially across individual mutations and peptide positions. No model performed best throughout the antigen sequence. Mutations at the experimentally restrictive R5 position were consistently more difficult to predict than mutations at other positions. More generally, cross-model prediction performance declined as mutations produced stronger or broader experimental losses of TCR recognition, with Spearman correlations of $\rho = -0.723$ for recognition disruption and $\rho = -0.876$ for binder-loss fraction. These results reveal a systematic failure mode not captured by conventional pairwise benchmarks: current models are least reliable for antigen mutations with the largest experimentally observed effects on TCR recognition.

Availability and Implementation: Code and processed benchmark data will be made publicly available upon publication.

1 Introduction

T cell receptors (TCRs) recognize antigenic peptides presented by major histocompatibility complex (MHC) molecules and thereby mediate antigen-specific cellular immune responses [5]. Characterizing TCR–peptide–MHC (pMHC) recognition is important for understanding immune responses to infection, vaccination, cancer, and autoimmunity, as well as for the development of T-cell-based diagnostics and therapeutics. Experimental methods such as pMHC multimer staining provide direct measurements of antigen-specific T cells [3], while more recent high-throughput approaches enable multiplexed mapping of TCR specificity across large antigen panels [18, 8]. Nevertheless, exhaustive experimental characterization remains difficult because of the enormous combinatorial space of TCR and antigen sequences.

Computational TCR–epitope prediction has therefore emerged as an important complement to experimental characterization. Over the past decade, a broad range of sequence-based and machine-learning approaches have been developed to predict TCR antigen specificity, using representations ranging from CDR3-based inputs to paired TCR $\alpha\beta$ chains, additional CDR loops, gene usage, and HLA context. Recent community benchmarking efforts such as IMMREP22 and IMMREP23 have evaluated many of these approaches under common test settings and highlighted both progress and substantial remaining limitations in TCR–epitope prediction [15, 12].

Despite continued methodological development, reliable generalization remains a major challenge. Previous studies have shown that TCR-binding predictors can exhibit substantial performance degradation when evaluated on previously unseen peptides [6]. The IMMREP23 community benchmark similarly demonstrated

the difficulty of generalizing to new epitopes [15]. Independent comparisons of TCR–pMHC prediction methods have further shown that performance can depend strongly on training data and evaluation-set composition [9]. Together, these studies demonstrate that strong performance on conventional TCR–epitope benchmarks does not necessarily imply robust generalization to new antigen-recognition settings.

Most existing evaluations nevertheless formulate TCR specificity primarily as a pairwise discrimination problem: given a TCR and an epitope, the model is evaluated on whether it distinguishes experimentally positive from negative interactions. This formulation does not directly address an important biological use case. Antigen sequences vary and evolve, and even a single amino-acid substitution can substantially alter TCR recognition. Because TCRs must recognize related but non-identical peptide sequences, cross-reactivity is an intrinsic property of cellular immunity [16]. At the same time, antigen substitutions can alter or eliminate recognition by particular T-cell populations. Such effects have been observed for SARS-CoV-2 variants, where mutations can differentially affect CD8⁺ T-cell responses [2], and cross-reactive T-cell recognition has been an important consideration in understanding immunity to SARS-CoV-2 [19].

Importantly, the effects of antigen mutations are structured rather than uniform. Some peptide positions tolerate broad amino-acid variation, whereas others are strongly constrained, and mutations at the same position can affect different TCRs differently. This raises a complementary model-evaluation question that is not captured by conventional pairwise benchmarks: *can current TCR–epitope prediction models reproduce experimentally observed changes in TCR recognition caused by antigen mutations?* Answering this question requires an evaluation design in which mutations are systematically related to a common reference antigen and both retained and lost recognition are measured experimentally.

The TCR Fingerprinting study of Malone et al. [10] provides such an opportunity. The study systematically characterized recognition of the HLA-A*02:01-restricted SARS-CoV-2 epitope YLQPRTFLL (YLQ) and its single-amino-acid variants. The experimental panel contains the reference peptide together with all 171 possible single-amino-acid substitutions across its nine positions. Twenty-one TCRs were evaluated against this peptide panel using background-normalized pMHC staining, producing quantitative recognition measurements and experimentally defined binding outcomes. The resulting data reveal a highly structured mutation landscape in which some positions are broadly tolerant, others produce greater recognition losses, and the central R5 residue is particularly restrictive [10].

These data differ fundamentally from benchmark collections assembled from unrelated TCR–epitope pairs. Because every mutant peptide represents a controlled perturbation of the same reference antigen, the dataset enables prediction performance to be evaluated at the level of individual mutations and antigen positions. Moreover, non-binding interactions are experimentally observed rather than generated through random re-assignment of TCRs and peptides. This provides an opportunity to examine not only whether models distinguish binding from non-binding interactions, but also where in the mutation landscape current models succeed or fail.

In this study, we construct a mutation-aware computational benchmark from the Fingerprinting measurements and use it to evaluate eight representative TCR–epitope prediction models. The evaluated methods span different model architectures, TCR input representations, and training-data sources. We use a model-independent evaluation procedure in which prediction performance is computed separately for individual antigen mutations and then summarized across mutations and peptide positions. Because the evaluated models were trained using different public resources, we additionally examine model-specific training–benchmark overlap before evaluation. We have found that overlap with the Fingerprinting panel does not exist for most models and is very limited for one model.

Our evaluation reveals several consistent limitations of current prediction methods. Overall mutation-level discrimination is modest, with the best model achieving a Macro AUC_{0.1} below 0.65. Prediction performance also varies substantially across individual mutations and peptide positions, and no single model performs best throughout the antigen sequence. In particular, substitutions at the experimentally restrictive R5 position are more difficult to predict than mutations at the remaining positions. More broadly, prediction performance decreases systematically as mutations produce stronger or more widespread experimental losses of TCR recognition. Thus, the mutations with the largest experimentally observed effects are also among the most challenging for current computational models.

The main contributions of this study are threefold:

- We construct a mutation-aware computational benchmark from systematic experimental perturbations of a common reference antigen. Because all test peptides differ from the same epitope by a single amino-acid substitution, the benchmark enables controlled evaluation of TCR–epitope recognition across a densely sampled local mutation landscape of 171 antigen variants.
- We introduce a mutation-resolved evaluation framework that exposes where current predictors succeed or fail within a controlled antigen landscape, including mutation-specific, position-specific, and R5-specific behavior that is hidden by aggregate TCR–epitope classification metrics.
- Through systematic evaluation of eight representative prediction models, we identify a consistent failure mode of current methods: prediction performance is strongly mutation dependent and deteriorates for antigen substitutions that produce larger or broader experimentally observed losses of TCR recognition.

Together, these results extend evaluation of TCR specificity prediction beyond the conventional question of whether a TCR recognizes an antigen toward the more demanding question of whether computational models can predict how antigen sequence variation changes TCR recognition.

2 Materials and Methods

2.1 Overview of the Mutation-Aware Benchmarking Framework

This study does not propose a new T cell receptor (TCR)–epitope prediction model. Instead, we develop a model-independent benchmarking framework to evaluate whether existing computational models reproduce experimentally observed effects of antigen mutations on TCR recognition. The overall workflow is illustrated in Fig. 1.

The framework consists of five major components: (1) construction of a computational benchmark from systematic single-amino-acid mutagenesis experiments; (2) examination of the training data used by individual prediction models to identify potential training–benchmark overlap; (3) inference of TCR–epitope prediction scores using the original pretrained models; (4) mutation- and position-level evaluation using experimentally measured binding outcomes; and (5) analysis of whether model performance deteriorates systematically for mutations that produce stronger experimentally observed disruption of TCR recognition.

Rather than treating all TCR–peptide interactions as independent test examples, the evaluation preserves the structured relationship among peptide variants derived from the same reference antigen. This design enables model performance to be examined at the level of individual antigen mutations, peptide positions, and biologically consequential mutation classes.

2.2 Experimental Fingerprinting Dataset

2.2.1 TCRs and antigen mutation panel

The experimental data used in this study were generated using the TCR fingerprinting approach reported by Malone et al. [10]. The study characterized recognition of the immunodominant SARS-CoV-2 epitope YLPRTFLL (YLQ), presented by HLA-A*02:01, together with a systematic panel of single-amino-acid substitutions.

The complete panel contains 172 peptide species: the reference YLPRTFLL peptide and 171 single-amino-acid variants corresponding to all 19 possible non-reference amino acids at each of the nine peptide positions. These peptides were evaluated against 21 TCRs. Twenty TCRs were identified using the reference YLQ peptide, whereas one additional TCR was identified using a pool of R5-mutant peptides.

The systematic mutation design differs from conventional TCR–epitope datasets assembled from unrelated positive and negative interactions. Each mutant differs from the reference antigen at exactly one defined residue, allowing the effect of antigen sequence perturbation on TCR recognition to be examined in a controlled mutation landscape.

Overview of the mutation-aware benchmarking framework

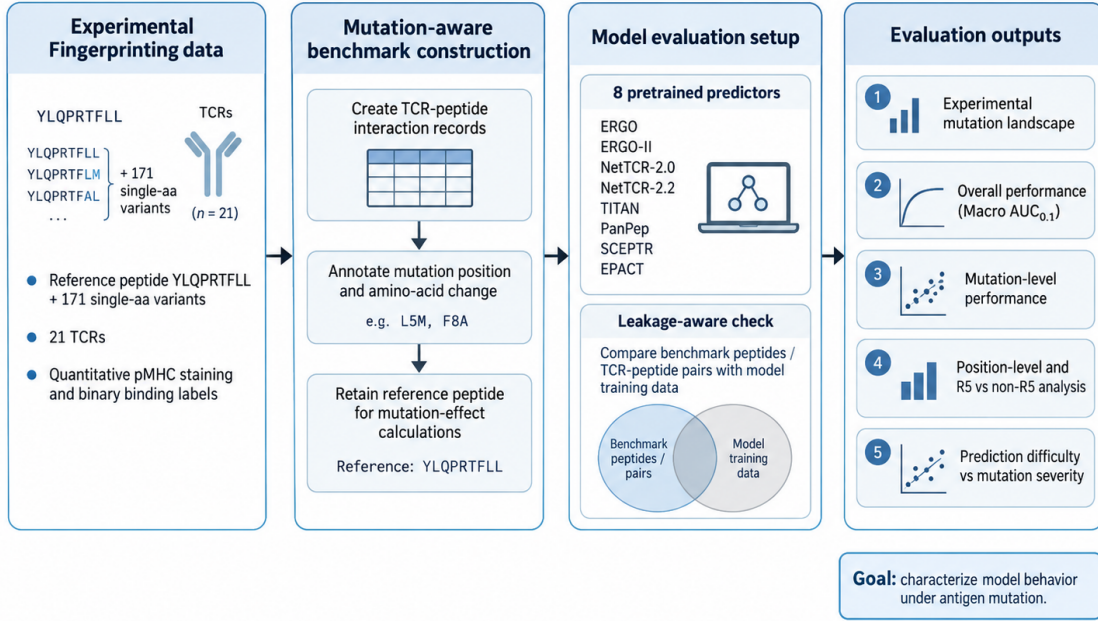


Figure 1: Overview of the mutation-aware benchmarking framework. The workflow starts from the experimental Fingerprinting dataset and model training data, performs overlap screening and model inference, and then evaluates prediction performance at the mutation, position, and mutation-severity levels.

2.2.2 Experimental pMHC-recognition measurements

TCR-specific peptide–MHC (pMHC) recognition was measured by staining TCR-transduced J76 cells with individual pMHC tetramers. Background staining was measured using TCR-negative J76 cells. For TCR i and peptide j , the quantitative experimental recognition signal was represented as

$$B_{ij} = \log_2 \left(\frac{\text{MFI}_{\text{TCR}^+, ij}}{\text{MFI}_{\text{TCR}^-, j}} \right), \quad (1)$$

where $\text{MFI}_{\text{TCR}^+, ij}$ denotes the pMHC staining intensity measured for TCR-transduced cells and $\text{MFI}_{\text{TCR}^-, j}$ denotes the corresponding background staining intensity measured using TCR-negative cells for peptide j .

Following the experimental definition used in the original study, TCR–pMHC interactions were classified as binders when

$$y_{ij} = \begin{cases} 1, & B_{ij} > 1, \\ 0, & B_{ij} \leq 1. \end{cases} \quad (2)$$

Thus, the Fingerprinting dataset provides two complementary experimental endpoints: a continuous background-normalized pMHC-recognition measurement and a binary indicator of experimentally retained binding.

The original study also evaluated the HLA-A*02:01 binding capacity of the mutant peptides. Peptides were assessed using NetMHCpan 4.1, with predicted strong and weak MHC binders defined by %Rank < 0.5

and %Rank < 2, respectively. Peptide–MHC stability was additionally evaluated experimentally using a T2 stabilization assay. These controls reduce the possibility that apparent changes in TCR recognition are driven primarily by failure of peptide presentation rather than altered TCR recognition.

2.3 Construction of the Fingerprinting Computational Benchmark

The Fingerprinting measurements were converted into a structured mutation-aware computational benchmark. Each benchmark record corresponds to a specific TCR and peptide species and contains the TCR sequence information, peptide sequence, mutation position, reference and substituted amino acids, quantitative experimental recognition measurement, and binary experimental binding label.

For mutant peptide j and TCR i , a benchmark interaction can be represented as

$$D_{ij} = (T_i, P_j, k_j, a_j^{\text{ref}}, a_j^{\text{mut}}, B_{ij}, y_{ij}), \quad (3)$$

where T_i denotes the TCR sequence information, P_j denotes the mutant peptide sequence, k_j is the mutation position, a_j^{ref} and a_j^{mut} are the reference and substituted amino acids, respectively, and B_{ij} and y_{ij} are the continuous and binary experimental measurements.

The unmutated YLQPRTFLL peptide was retained as the experimental reference for mutation-effect calculations but was excluded from mutation-level prediction evaluation. The resulting mutation benchmark therefore contains 171 distinct single-amino-acid substitutions and 3,591 TCR–mutant interactions.

2.4 Evaluated TCR–Epitope Prediction Models

We evaluated eight existing TCR–epitope prediction models: ERGO [7], ERGO-II [17], NetTCR-2.0 [13], NetTCR-2.2 [11], TITAN [1], PanPep [4], SCEPTR [14], and EPACT [20].

These models were selected to provide a diverse cross-section of recent TCR-specificity predictors that are publicly available and can be applied to the Fingerprinting dataset without retraining. Collectively, they span different generations of model development, TCR input representations, training-data sources, and modeling strategies, ranging from relatively compact CDR3-based predictors to models that incorporate paired $\alpha\beta$ chains, additional CDR loops, gene usage, and HLA context. The purpose of this selection is not to establish an exhaustive ranking of all published TCR–epitope predictors, but to assess whether current representative methods with substantially different designs exhibit similar strengths and limitations when evaluated on a controlled antigen-mutation benchmark. The Fingerprinting dataset provides a superset of the sequence and annotation information required by these models, including paired TCR α/β sequences, CDR information, V/J gene annotations, peptide sequence, and the HLA-A*02:01 restriction. For each predictor, we therefore extracted the subset of benchmark fields required by its published input representation. This allows all models to be evaluated on the same underlying experimental interactions while preserving their original input specifications.

The evaluated models differ substantially in the amount of TCR and pMHC information incorporated during prediction. ERGO uses CDR3 β and peptide sequence, whereas ERGO-II can additionally incorporate paired CDR3 α , V/J gene usage, MHC allele, and T-cell type. NetTCR-2.0 uses CDR3 α and/or CDR3 β together with peptide sequence, while NetTCR-2.2 incorporates all six CDR1–3 loops from the paired α and β chains. TITAN and PanPep primarily use TCR β sequence information together with the peptide sequence. In contrast, SCEPTR and EPACT use richer paired $\alpha\beta$ TCR representations and jointly model the peptide together with HLA information.

The models were originally developed using different combinations of public and experimentally generated TCR–epitope datasets, including McPAS-TCR, VDJdb, IEDB, PIRD, TBAdB, 10x Genomics multimer data, and other curated cohorts. These differences in training data motivate the model-specific training–benchmark overlap analysis described in the following subsection.

For each model, we used the publicly released pretrained implementation and followed the corresponding inference procedure. All evaluated models produced a continuous binding score in the range [0, 1], with

larger values indicating stronger predicted TCR–epitope binding. Because the internal transformation used to obtain these scores is model-specific, we treated the released model score as the prediction output without additional calibration or rescaling. The continuous scores were used directly for ROC-based evaluation and were not converted to binary predictions using model-specific thresholds.

Table 1 summarizes the input fields, training data, output representation, and implementation information for the eight evaluated models.

Model	TCR input used	Peptide input	MHC input	Training data	Version
ERGO	CDR3 β	Epitope sequence	Not used	McPAS-TCR and VD-Jdb; additional tumor-derived data	
ERGO-II	CDR3 β with optional CDR3 α ; V/J gene usage and T-cell type can also be incorporated	Epitope sequence	MHC allele can be included	McPAS-TCR and VD-Jdb; 10x data excluded during training-data construction	
NetTCR-2.0	CDR3 α and/or CDR3 β , with paired chains used when available	Epitope sequence	Not explicitly modeled	IEDB and VDJdb positive interactions; 10x data used for negative examples	
NetTCR-2.2	CDR1–3 loops from both TCR α and TCR β chains	Epitope sequence	Not explicitly modeled	Curated IEDB and VD-Jdb binders; denoised 10x data used for negative examples	
TITAN	CDR3 β or TCR β sequence representation	Epitope sequence or molecular representation	Not used	Curated TCR–epitope interactions compiled primarily from McPAS-TCR and VDJdb-related sources	
PanPep	CDR3 β	Epitope sequence	Not explicitly modeled	IEDB, VDJdb, PIRD, McPAS-TCR, and additional SARS-CoV-2 cohort data	
SCEPTR	Paired TCR $\alpha\beta$ using all six CDR loops	Epitope sequence	HLA jointly modeled with peptide	Pretraining on approximately 842,000 paired $\alpha\beta$ clonotypes; downstream TCR–pMHC training/evaluation using curated antigen-specific data	
EPACT	Paired TCR $\alpha\beta$	Epitope sequence	HLA class I jointly modeled	IEDB, VDJdb, McPAS-TCR, TBAdb, SARS-CoV-2 datasets, and 10x multimer data	

Table 1: TCR–epitope prediction models evaluated in this study. The Fingerprinting dataset provides the sequence and annotation information required by all models, and each predictor was supplied with the subset of fields corresponding to its published input representation. All models produced continuous binding scores in the range $[0, 1]$, which were used directly for ROC-based evaluation without additional calibration or model-specific thresholding. The exact software or checkpoint version used for each model will be added after implementation verification.

2.5 Model-Specific Training–Benchmark Overlap Control

Potential training–benchmark overlap was examined separately for each model because the evaluated predictors were developed using different combinations of public TCR databases and experimental datasets. Original training datasets released by the model developers were used whenever available. When exact training sets were not publicly released, they were reconstructed from the data sources and preprocessing procedures described in the corresponding publications.

We distinguished several forms of prior exposure that may affect benchmark interpretation, including overlap at the peptide level and overlap involving specific TCR–peptide combinations. Peptide overlap was defined as the presence of an identical benchmark peptide sequence in the model’s training data. We additionally examined exact CDR3 β –peptide matches to identify benchmark interactions that may have been directly represented during model development.

Overall, training–benchmark overlap was minimal. For ERGO, ERGO-II, NetTCR-2.0, TITAN, EPACT,

Table 2: Training–benchmark overlap between the Fingerprinting dataset and the training data used by the evaluated prediction models.

Model	Peptide overlap	Rows with peptide overlap	Exact CDR3 β –peptide rows
ERGO	0/172 (0.0%)	0 (0.0%)	0 (0.0%)
ERGO-II	0/172 (0.0%)	0 (0.0%)	0 (0.0%)
NetTCR-2.0	0/172 (0.0%)	0 (0.0%)	0 (0.0%)
NetTCR-2.2	1/172 (0.6%)	21 (0.6%)	2 (0.1%)
TITAN	0/172 (0.0%)	0 (0.0%)	0 (0.0%)
EPACT	0/172 (0.0%)	0 (0.0%)	0 (0.0%)
PanPep	0/172 (0.0%)	0 (0.0%)	0 (0.0%)
SCEPTR	0/172 (0.0%)	0 (0.0%)	0 (0.0%)

PanPep, and SCEPTR, none of the 172 Fingerprinting peptides were identified in the corresponding training data, and no exact CDR3 β –peptide matches were observed. NetTCR-2.2 showed only limited overlap: one of the 172 peptide species (0.6%) was present in its training data, corresponding to 21 benchmark rows (0.6% of interactions), and only two benchmark rows (0.1%) matched an exact CDR3 β –peptide combination.

These results indicate that the Fingerprinting benchmark is largely independent of the training data used by the evaluated models. The extremely limited overlap observed for NetTCR-2.2 was recorded explicitly and taken into account during leakage-aware evaluation.

Importantly, prior exposure to the reference YLQ sequence was distinguished from direct exposure to individual mutant interactions. Because the benchmark is designed to evaluate the effects of antigen perturbation around a common reference sequence, knowledge of the reference antigen does not necessarily constitute the same form of leakage as direct exposure to a benchmark mutant or an identical TCR–mutant pair.

2.6 Experimental Characterization of the Mutation Landscape

Before evaluating computational predictions, we summarized the experimentally observed mutation landscape to establish the biological reference against which model performance was interpreted.

For each mutation j , the experimentally observed binder-retention fraction was calculated as

$$R_j = \frac{1}{N_j} \sum_{i=1}^{N_j} y_{ij}, \quad (4)$$

where N_j is the number of TCRs evaluated against mutation j and y_{ij} is the experimentally defined binary binding label.

Position-level experimental tolerance was then quantified by averaging R_j over the 19 possible substitutions occurring at each peptide position. At position P1, all 19 single-amino-acid substitutions retained experimentally defined binding across the tested TCRs, corresponding to a position-level mean binder-retention fraction of 1.0. P1 therefore represents a highly mutation-tolerant position in the experimental recognition landscape. This analysis was used to identify relatively mutation-tolerant and mutation-restrictive regions of the YLQPRTFLL epitope.

Because substitutions at the central R5 residue exhibited particularly strong recognition losses in the experimental data, binder-retention fractions for R5 substitutions were additionally compared with those for mutations at all remaining peptide positions (excluding P1).

2.7 Model Inference and Mutation-Level Prediction Evaluation

For model m , let $S_{ij}^{(m)}$ denote the continuous prediction score assigned to TCR i and mutant peptide j . Prediction performance was evaluated using the standardized partial area under the receiver operating characteristic curve over the low-false-positive-rate region $\text{FPR} \leq 0.1$, denoted $\text{AUC}_{0.1}$.

$AUC_{0.1}$ was calculated using the continuous model prediction scores and the experimentally defined binary labels. The standardized partial-AUC formulation was used such that a random classifier has an expected value of 0.5 and perfect discrimination has a value of 1.0. In implementation, this corresponds to the standardized partial ROC-AUC computed with `roc_auc_score(..., max_fpr=0.1)` in scikit-learn.

The fundamental evaluation unit was an individual antigen mutation. For each mutation j , $AUC_{0.1}$ was calculated across the TCRs evaluated against that mutant peptide:

$$A_j^{(m)} = AUC_{0.1} \left(\{y_{ij}\}_{i=1}^{N_j}, \{S_{ij}^{(m)}\}_{i=1}^{N_j} \right). \quad (5)$$

ROC-based evaluation requires both experimentally positive and negative interactions. Mutations for which all tested TCRs belonged to a single experimental class were therefore excluded from $AUC_{0.1}$ analysis. Among the 171 mutations, 113 contained both experimental classes and were AUC-eligible; the remaining 58 mutations were excluded from ROC-based evaluation.

Overall model performance was summarized using Macro $AUC_{0.1}$:

$$\text{Macro}AUC_{0.1}^{(m)} = \frac{1}{M} \sum_{j=1}^M A_j^{(m)}, \quad (6)$$

where $M = 113$ is the number of AUC-eligible mutations. This formulation gives equal weight to each mutation regardless of the number or class composition of its TCR interactions.

2.8 Position-Specific Prediction Performance

To determine whether prediction difficulty varies systematically across the antigen sequence, mutation-specific $AUC_{0.1}$ values were grouped according to the position of the substituted amino acid.

For model m and peptide position k , position-level performance was calculated as

$$A_k^{(m)} = \frac{1}{|M_k|} \sum_{j \in M_k} A_j^{(m)}, \quad (7)$$

where M_k denotes the set of AUC-eligible mutations occurring at position k . Thus, position-level performance represents the mean discrimination performance across individual substitutions at a position rather than a single ROC curve obtained by pooling all TCR–mutation interactions.

P1 was excluded from position-level ROC analysis because none of its substitutions contained both experimentally positive and negative TCR interactions. Experimentally, P1 substitutions were uniformly well tolerated: all tested TCR–mutant interactions remained above the binding threshold, yielding a mean binder-retention fraction of 1.0 at this position. Consequently, P1 provides an informative biological observation but does not support discrimination-based evaluation with $AUC_{0.1}$, which requires both positive and negative classes.

To further examine the experimentally identified importance of the central R5 residue, we compared mean mutation-specific $AUC_{0.1}$ for R5 substitutions with that for mutations at all other positions. The R5 group contained 19 AUC-eligible substitutions and the non-R5 group contained 94.

2.9 Mutation Severity and Prediction Difficulty

The position-level analysis identifies where prediction models perform well or poorly, but does not determine whether prediction difficulty is related to the biological impact of an individual mutation. We therefore examined whether mutations producing stronger experimentally observed perturbations of TCR recognition are systematically more difficult for the evaluated models.

We characterized mutation severity using two complementary experimental measures. The first quantifies the magnitude of the mutation-induced change in TCR recognition. For TCR i and mutant peptide j , the change relative to the reference YLQPRTFLL peptide was defined as

$$\Delta B_{ij} = B_{ij} - B_{i,\text{ref}}, \quad (8)$$

where $B_{i,\text{ref}}$ is the experimental $\log_2$ fold-change measured for the same TCR against the reference peptide. The experimental disruption score for mutation j was then defined as

$$D_j = -\frac{1}{N_j} \sum_{i=1}^{N_j} \Delta B_{ij}. \quad (9)$$

Larger positive values of D_j therefore indicate a greater average loss of TCR recognition relative to the reference peptide.

The second measure characterizes the breadth of recognition loss across TCRs. For mutation j , the binder-loss fraction was defined as

$$L_j = 1 - \frac{1}{N_j} \sum_{i=1}^{N_j} y_{ij}. \quad (10)$$

Thus, $L_j = 0$ indicates that all tested TCRs retain experimentally defined binding, whereas larger values indicate that the mutation causes loss of binding across a greater fraction of TCRs.

To characterize prediction difficulty independently of any single model, we averaged mutation-specific $\text{AUC}_{0.1}$ across the eight evaluated models:

$$\bar{A}_j = \frac{1}{K} \sum_{m=1}^K A_j^{(m)}, \quad (11)$$

where $K = 8$.

Spearman’s rank correlation coefficient was then used to assess the relationship between prediction performance $\bar{A}_j$ and each experimental severity measure, D_j and L_j , across the 113 AUC-eligible mutations. This analysis tests whether the mutations that produce the strongest or broadest experimentally observed disruption of TCR recognition also represent systematic failure modes for current prediction models.

2.10 Statistical Analysis

Standardized $\text{AUC}_{0.1}$ was used as the primary prediction-performance metric because it evaluates discrimination within the low-false-positive-rate region while avoiding dependence on model-specific decision thresholds. Mutation-level $\text{AUC}_{0.1}$ values were treated as the fundamental units for macro averaging and position-level aggregation.

Spearman’s rank correlation coefficient was used to evaluate associations between experimentally measured mutation severity and cross-model prediction performance because these relationships were not assumed to be linear. All reported correlation tests were two-sided.

Analyses were implemented in Python using pandas, NumPy, SciPy, and scikit-learn. ROC-based metrics were calculated using `sklearn.metrics.roc_auc_score` with `max_fpr=0.1`.

3 Results

3.1 Experimental mutation landscape used as the evaluation reference

Before evaluating computational predictions, we first summarized the experimentally observed mutation landscape of the YLQPRTFLL epitope to establish the reference patterns that a mutation-aware prediction model should recover. The Fingerprinting measurements reveal substantial structure across both peptide positions and individual TCRs (Fig. 2a). Some substitutions preserve recognition across most of the tested receptors, whereas others produce marked losses of recognition. Importantly, these effects are not uniform across TCRs, indicating that the consequences of a given antigen mutation depend strongly on receptor context.

At the position level, mutations at P1 and P2 were broadly tolerated, with mean binder-retention fractions of 1.000 and 0.982, respectively (Fig. 2b). P8 and P9 were also relatively tolerant, with mean binder fractions of 0.887 and 0.937. In contrast, substitutions at several internal positions produced larger recognition losses. Mean binder fractions were 0.855 at P3 and P4, 0.734 at P6, and 0.767 at P7.

The strongest positional restriction was observed at P5. Across the 19 possible substitutions of the reference arginine residue, the mean binder-retention fraction decreased to 0.569, the lowest among all peptide positions. Consistently, R5 substitutions showed substantially lower overall binder retention than mutations at the remaining positions (0.569 versus 0.877; Fig. 2c).

These observations define the experimental reference for the subsequent model evaluations. In particular, a useful predictor should not only distinguish binding from non-binding interactions in aggregate, but should also reflect the broad positional tolerance pattern, the unusually restrictive behavior of R5 substitutions, and the substantial mutation-specific variation observed across TCRs. We therefore use these experimentally measured patterns as the basis for interpreting mutation- and position-resolved prediction performance.

3.2 Overall model performance on the mutation benchmark

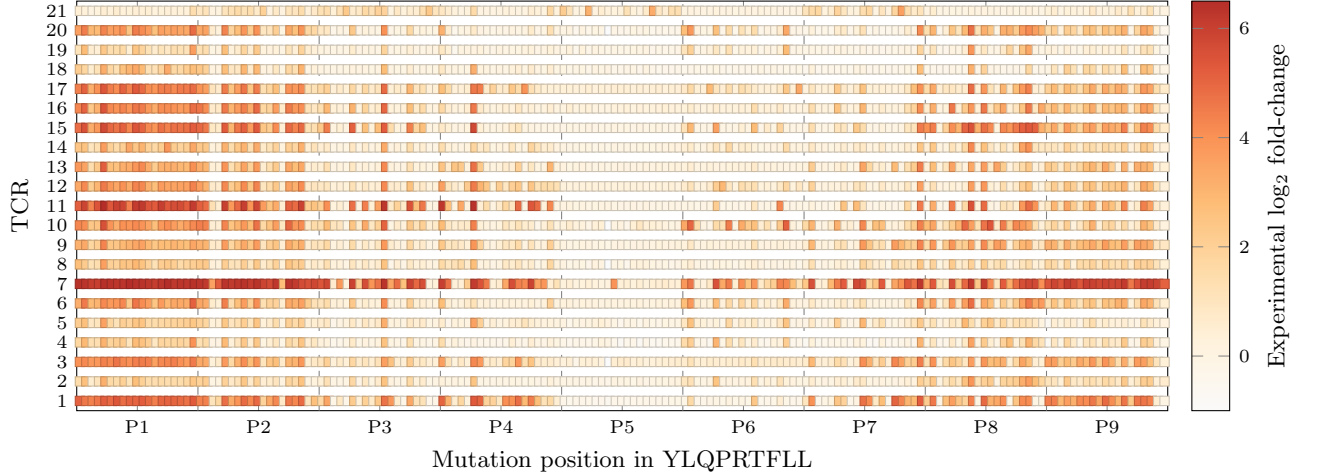
We first evaluated the overall ability of the eight TCR–epitope prediction models to distinguish experimentally observed binding and non-binding interactions across the mutation benchmark. After excluding the unmutated reference peptide, the benchmark contained 3,591 TCR–peptide interactions corresponding to 171 single-amino-acid variants of YLQPRTFLL. Among these interactions, 3,027 were experimentally classified as binders and 564 as non-binders according to the experimentally defined $\log_2$ fold-change criterion.

Because our objective is to evaluate model performance across individual antigen mutations, we used Macro $AUC_{0.1}$ as the primary overall performance measure. Following the low-false-positive-rate evaluation used in IMMREP23, $AUC_{0.1}$ was calculated separately for each peptide variant across the tested TCRs and then averaged across variants. Of the 171 mutations, 113 contained both experimentally positive and negative TCR interactions and therefore supported ROC-based evaluation. The remaining mutations contained only a single experimental class and were excluded from the $AUC_{0.1}$ calculation.

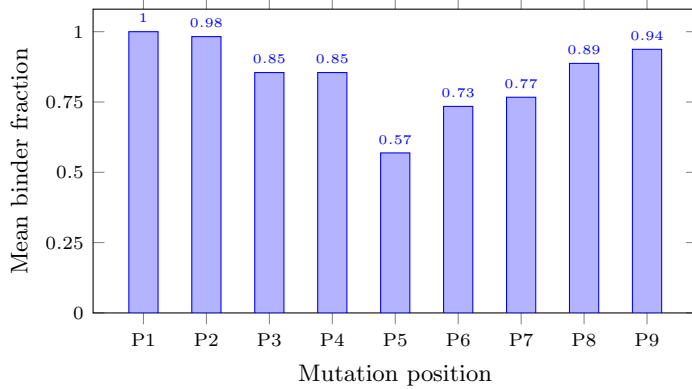
As shown in Fig. 3, overall predictive performance was modest for all eight models. ERGO achieved the highest Macro $AUC_{0.1}$ of 0.644, whereas ERGO2 and SCEPTR showed lower performance, with Macro $AUC_{0.1}$ values of 0.553 and 0.533, respectively.

Thus, although several models performed above the random-reference level of 0.5, even the best-performing model remained below 0.65. These results indicate that current TCR–epitope prediction models provide only limited overall discrimination across experimentally characterized antigen mutations.

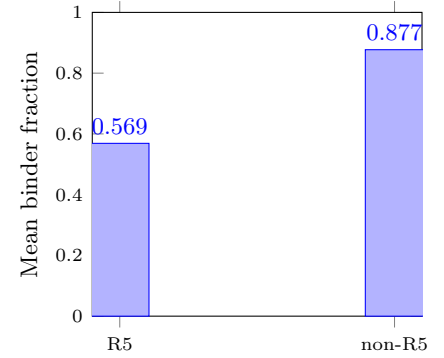
The macro-average summarizes performance across mutation contexts but does not reveal whether prediction quality is consistent across individual substitutions or peptide positions. We therefore next examine mutation- and position-specific performance to identify where individual models succeed or fail within the experimental mutation landscape.



(a) Experimental TCR recognition fingerprints across all 171 single-amino-acid substitutions of YLQPRTFLL. Color indicates the background-normalized pMHC staining response measured as $\log_2(\text{signal MFI}/\text{background MFI})$.



(b) Position-wise experimental tolerance. Binder fraction is the proportion of tested TCRs with $\log_2$ fold-change > 1 , averaged over the 19 substitutions at each position.



(c) Focused comparison of substitutions at the central R5 residue with mutations at all other peptide positions.

Figure 2: Experimentally observed mutation landscape of the YLQPRTFLL epitope. (a) Quantitative TCR recognition fingerprints measured by background-normalized pMHC staining across 21 TCRs and the 171 possible single-amino-acid substitutions. (b) Mean binder-retention fraction across mutations at each peptide position, using the experimentally defined $\log_2$ fold-change > 1 criterion. Substitutions at P1 and P2 are broadly tolerated, whereas internal positions show stronger recognition losses, with P5 exhibiting the lowest overall retention. (c) R5 substitutions retain TCR recognition substantially less frequently than substitutions at other positions. Together, these measurements establish both a pronounced position-dependent mutation landscape and substantial TCR-specific variation, which provide the experimental reference for the subsequent model evaluations.

3.3 Mutation- and position-specific prediction performance

We next examined how prediction performance varied across individual mutations and peptide positions. Although the preceding analysis showed only modest overall performance, aggregate metrics can obscure substantial variation among specific substitutions. We therefore calculated $\text{AUC}_{0.1}$ separately for each of the 113 mutations that contained both experimentally positive and negative TCR interactions.

As shown in Fig. 4, mutation-specific performance varied considerably within every model. ERGO showed

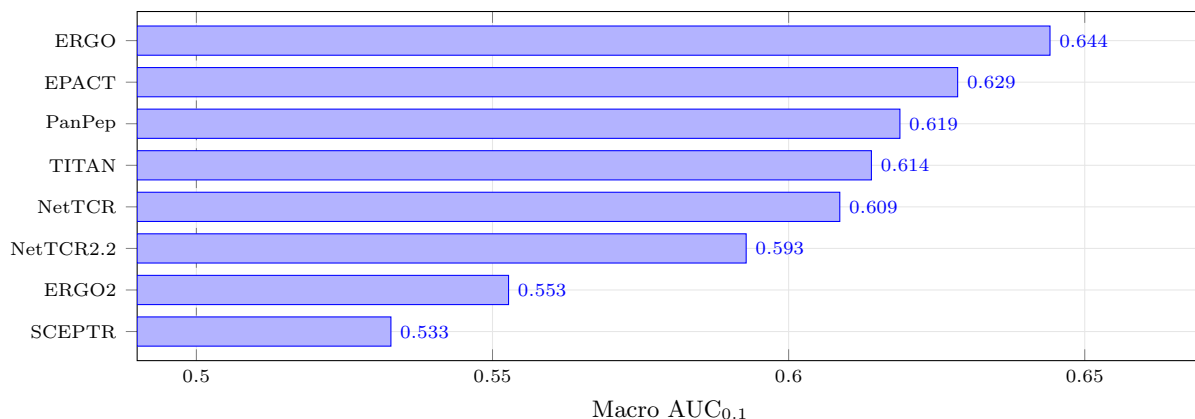


Figure 3: **Overall model performance on the mutation benchmark.** Macro AUC_{0.1} was calculated by first computing AUC_{0.1} separately for each peptide mutation and then averaging across the 113 mutations containing both experimentally positive and negative TCR interactions. ERGO achieved the highest overall performance, followed by EPACT and PanPep, although all models showed only modest discrimination. The dashed line indicates the random-reference value of 0.5.

the highest median mutation-specific AUC_{0.1}, whereas EPACT, NetTCR, PanPep, and TITAN also exhibited substantial subsets of mutations with relatively good discrimination. In contrast, ERGO2 and SCEPTR were concentrated much closer to the random-reference level. Importantly, even the stronger models included many individual mutations with AUC_{0.1} values near 0.5, while other mutations were predicted substantially better. These results indicate that prediction difficulty depends strongly on the specific amino-acid substitution rather than being uniform across the mutation landscape.

We then aggregated mutation-specific performance by peptide position to determine whether this variation exhibited positional structure (Fig. 5). Clear differences were observed across both positions and models. For example, PanPep performed best at P2 and P4, with mean AUC_{0.1} values of 0.796 and 0.731, respectively. Other models, including ERGO, TITAN, NetTCR2.2, and NetTCR also achieve the best performance at other positions. P1 is not included in this analysis because all P1 substitutions retained the same experimental binding class and therefore did not support ROC-based evaluation.

Notably, the model with the best overall Macro AUC_{0.1} was not the best-performing model at every peptide position. Instead, the leading model varied across the antigen sequence, indicating that different prediction methods capture different aspects of the mutation landscape. The position with the strongest experimentally observed recognition loss, P5, also remained particularly challenging from the prediction perspective: even the best-performing model at this position, TITAN, achieved a mean AUC_{0.1} of only 0.588. In contrast, substantially higher position-level performance was observed at several other positions, including P2, P4, P8, and P9.

Because the experimental analysis identified the central R5 residue as the most restrictive mutation site, we further compared model performance on R5 substitutions with performance on mutations at all other positions (Fig. 6). For all eight models, mean AUC_{0.1} was lower for R5 substitutions than for non-R5 mutations. For example, ERGO decreased from 0.660 on non-R5 mutations to 0.568 on R5 substitutions, EPACT from 0.652 to 0.514, PanPep from 0.642 to 0.506, and NetTCR2.2 from 0.614 to 0.489. TITAN, the best-performing model on R5 substitutions, achieved a mean AUC_{0.1} of only 0.588, compared with 0.619 on non-R5 mutations. Thus, the mutation class that is experimentally most disruptive is also consistently more difficult for current computational models to predict.

Together, these results show that aggregate model rankings conceal substantial mutation- and position-dependent differences in prediction quality. No single method consistently dominates across the YLQPRT-FLL sequence, and predictive performance remains especially limited for biologically important mutation contexts such as R5 substitutions.

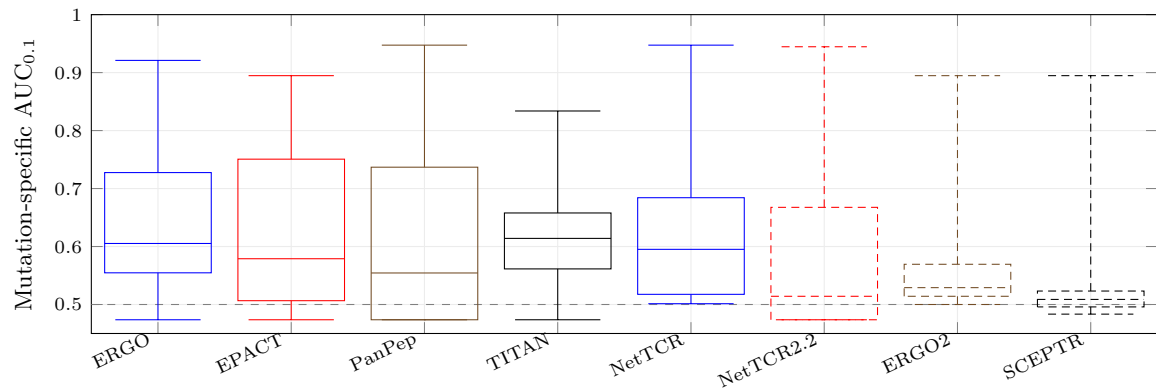


Figure 4: **Mutation-specific prediction performance.** Distribution of mutation-specific $AUC_{0.1}$ values across the 113 variants containing both experimentally positive and negative TCR interactions. Prediction performance varies substantially across individual substitutions for all eight models. The dashed line indicates the random-reference level of 0.5.

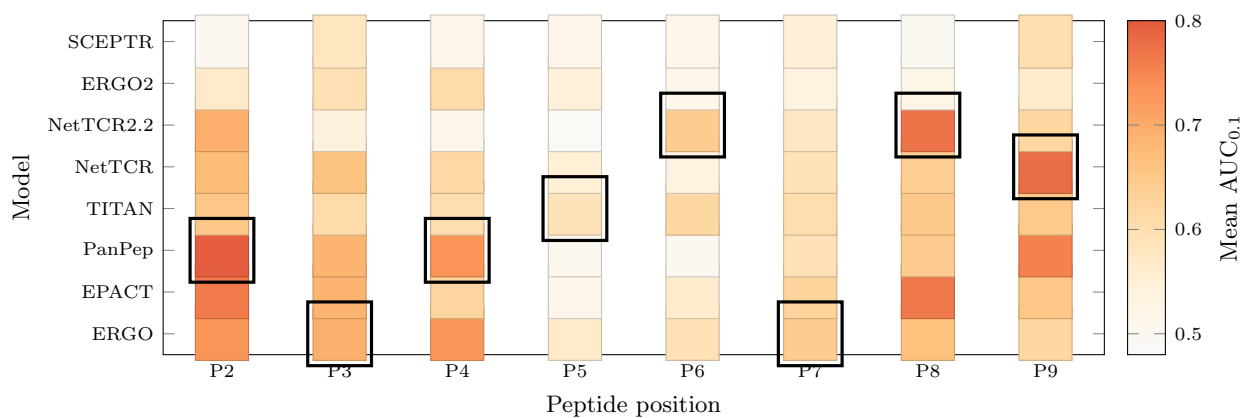


Figure 5: **Position-specific prediction performance.** Mean mutation-specific $AUC_{0.1}$ for each model at each peptide position. Each value averages the $AUC_{0.1}$ values of all AUC-eligible substitutions at that position. Black outlines identify the highest-performing model at each position. P1 is omitted because none of its substitutions contained both experimental classes.

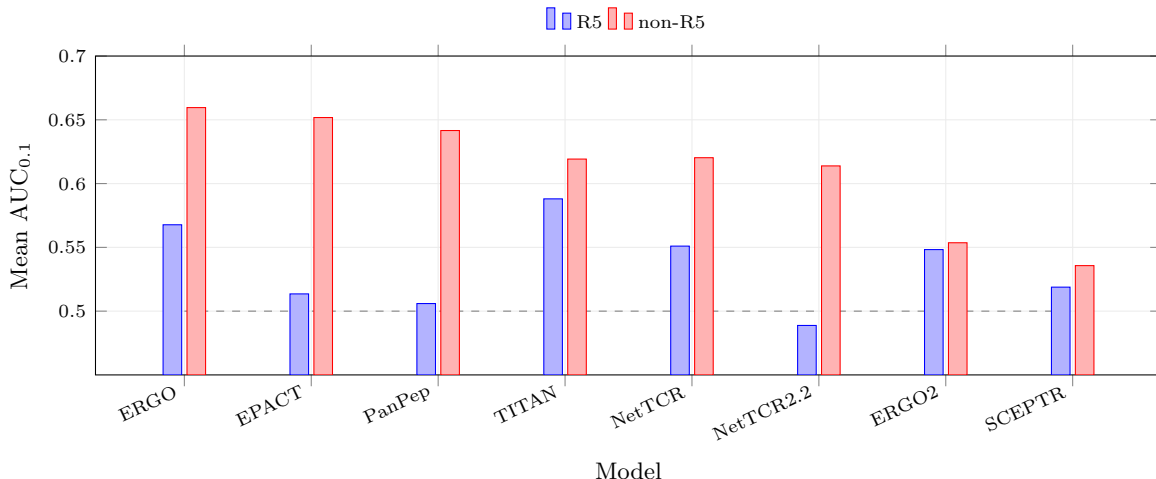


Figure 6: **Prediction performance for R5 and non-R5 mutations.** Mean AUC_{0.1} was calculated across the 19 R5 substitutions and the 94 AUC-eligible non-R5 substitutions, respectively. R5 mutations were more difficult to predict for all eight models. The dashed line indicates the random-reference level of 0.5.

3.4 Prediction difficulty increases with experimentally observed mutation severity

The preceding analyses showed that model performance varies substantially across individual mutations and peptide positions. We next examined whether this variation follows a systematic pattern that can help characterize the failure modes of current TCR–epitope prediction models. Specifically, we asked whether mutations that produce larger experimentally observed changes in TCR recognition are also more difficult for the tested models to predict.

To characterize mutation severity, we used two complementary experimental measures. The first captures the magnitude of the mutation effect and is defined as the mean decrease in $\log_2$ fold-change relative to the reference YLQPRTFLL peptide across the tested TCRs. The second captures the breadth of the effect and is defined as the binder-loss fraction, i.e., the proportion of tested TCRs that no longer satisfy the experimentally defined binding criterion for the mutant peptide.

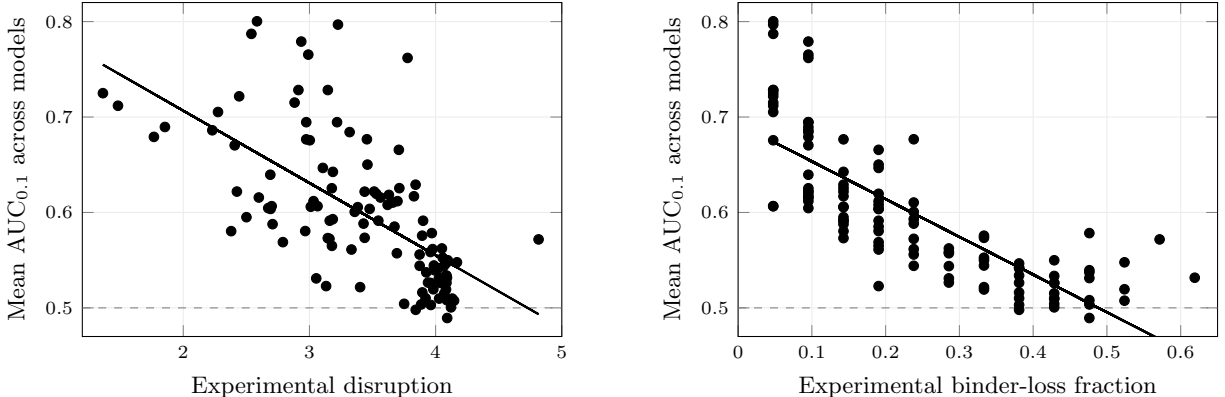
For each AUC-eligible mutation, we compared these measures with the mean mutation-specific AUC_{0.1} across the eight evaluated models. As shown in Fig. 7a, prediction performance decreased as the experimentally measured magnitude of recognition loss increased. Experimental disruption was negatively correlated with cross-model AUC_{0.1} (Spearman $\rho = -0.723$, $p = 1.6 \times 10^{-19}$).

A similar and even stronger pattern was observed for the binder-loss fraction (Fig. 7b). Mutations that caused loss of binding across a larger fraction of TCRs were associated with lower prediction performance (Spearman $\rho = -0.876$, $p = 6.7 \times 10^{-37}$).

These results characterize an important failure mode of the evaluated models. Prediction difficulty is not distributed uniformly across antigen mutations; rather, the models perform worst on mutations that produce the largest or broadest experimentally observed changes in TCR recognition. This extends the position-level findings by showing that the difficulty of R5 substitutions reflects a broader relationship between mutation severity and prediction performance across the mutation landscape.

4 Discussion

This study evaluates TCR–epitope prediction from a mutation-aware perspective rather than relying only on conventional pairwise binding classification. Using the experimentally characterized YLQPRTFLL mutation landscape, we found that current models provide only modest overall discrimination across antigen variants



(a) Relationship between continuous experimental mutation disruption and prediction performance. Experimental disruption is defined as the negative mean change in $\log_2$ fold-change relative to the reference peptide, such that larger values indicate stronger loss of TCR recognition. Each point represents one AUC-eligible mutation. Prediction performance is averaged across the eight evaluated models. Spearman $\rho = -0.723$ ($p = 1.6 \times 10^{-19}$).

(b) Relationship between experimentally observed binder loss and prediction performance. Binder-loss fraction is the proportion of tested TCRs that lose experimentally defined binding for a given mutation. Mutations causing greater loss of recognition are substantially more difficult for current prediction models. Spearman $\rho = -0.876$ ($p = 6.7 \times 10^{-37}$).

Figure 7: Experimentally disruptive antigen mutations are more difficult for current TCR-epitope prediction models. (a) Cross-model prediction performance decreases as the experimentally measured reduction in TCR recognition becomes larger. (b) An even stronger relationship is observed when mutation severity is measured by the fraction of TCRs that lose binding. Each point represents one of the 113 mutations supporting $AUC_{0.1}$ evaluation, and the y-axis shows the mean mutation-specific $AUC_{0.1}$ across the eight evaluated models. Dashed lines indicate the random-reference value of 0.5.

and that prediction performance varies substantially across both individual mutations and peptide positions. No single model consistently performed best across the antigen sequence, suggesting that current approaches capture different and incomplete aspects of TCR recognition.

The position-specific analysis further showed that model difficulty reflects biologically meaningful structure in the experimental landscape. In particular, substitutions at the central R5 residue, which produced the strongest experimental loss of recognition, were also more difficult to predict than mutations at other positions. More generally, prediction performance decreased as mutations produced stronger or broader experimentally observed disruption of TCR recognition. This result is important because it indicates that model errors are not uniformly distributed across mutation space. Instead, current predictors tend to perform worst on mutations that have the largest biological consequences for TCR recognition.

These findings complement recent benchmarking studies showing limited generalization of TCR-epitope prediction models to unseen or underrepresented antigen settings. Conventional benchmarks primarily ask whether a model can distinguish positive from negative TCR-epitope pairs. Our results suggest that even when models provide useful discrimination at the interaction level, this does not necessarily imply that they can accurately predict the consequences of antigen sequence variation. Mutation-aware evaluation therefore exposes limitations that may remain hidden in aggregate benchmark performance.

The results also have implications for future model development. Improved TCR-epitope predictors may need to capture not only sequence compatibility between a TCR and peptide, but also the structural and biochemical consequences of localized antigen perturbations. The substantial variation in model performance across positions suggests that current sequence representations do not fully encode positional constraints governing TCR recognition. Training and evaluation datasets containing systematic mutational perturbations may therefore be valuable for developing models that generalize more reliably to antigenic variation.

Several limitations should be noted. The benchmark is based on a single HLA-A*02:01-restricted SARS-

CoV-2 epitope and 21 TCRs, and the observed positional patterns may not generalize directly to other epitopes, HLA alleles, or TCR repertoires. In addition, although the systematic mutation panel provides dense coverage of local antigen sequence space, it evaluates only single-amino-acid substitutions. Future studies should extend mutation-aware benchmarking to additional antigens, HLA contexts, larger TCR repertoires, and combinations of mutations.

Overall, this work shows that mutation-aware evaluation provides information that is complementary to conventional TCR–epitope benchmarks. By linking prediction performance directly to experimentally measured antigen perturbations, the framework helps identify not only how well current models perform, but also which types of biologically consequential mutations remain difficult to predict.

5 Conclusion

We developed a mutation-aware benchmark for evaluating whether current TCR–epitope prediction models reproduce experimentally observed effects of single-amino-acid antigen variation. Across eight representative models, overall discrimination was modest, prediction performance varied substantially across mutations and peptide positions, and no single model performed best throughout the antigen sequence. Mutations at the experimentally restrictive R5 position were particularly difficult to predict, and more generally, prediction performance declined as mutations produced stronger or broader losses of TCR recognition.

These results show that conventional TCR–epitope binding performance does not fully characterize a model’s ability to predict the consequences of antigen sequence variation. Mutation-aware benchmarking provides a complementary evaluation perspective that can help identify biologically important failure modes and guide the development of models that generalize more reliably to antigenic variation.

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